(1) Compute the range of the following data :

710, 635, 423, 221, 971, 843, 307, 289.

Solution : Smallest Value = S = 221

Largest Value = L = 971

Range = L – S = 971 - 221 = 750

(2) Calculate the range of the following data :

Marks	0-10	10-20	20-30	30-40	40-50
No. of Students	4	17	22	16	11

Solution : L = Upper limit of highest class = 50

S = Lower limit of lowest class = 0

∴ Range = L – S = 50 – 0 = 50

(3) A die is rolled 30 times and the following distribution is obtained. Find the variance and the standard deviation.

Score	1	2	3	4	5	6
Frequency	2	6	2	5	9	6

Solution :

x	f	fx	fx^2
1	2	2	2
2	6	12	24
3	2	6	18
4	5	20	80
5	9	45	225
6	6	36	216
Total	30	121	565

$\bar{x} = \frac{1}{N} \Sigma fx = \frac{121}{30}$

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Now $\sigma x^2 = \frac{\Sigma f x^2}{N} - \bar{x}^2$

$= \frac{565}{30} - \left(\frac{121}{30}\right)^2 = \frac{565}{30} - \frac{14641}{900}$

$= \frac{16950 - 14641}{900} = \frac{2305}{900} = 2.57$

$S.D = \sqrt{\sigma^2 x} = \sqrt{2.57} = 1.60$

Variance $= 2.57$ S.D $= 1.60$

Repeat the experiment and construct the frequency table as shown above. Find its mean and standard deviation.

x	f	fx	fx^2
1	2	2	2
2	6	12	24
3	2	6	18
4	5	20	80
5	9	45	225
6	6	36	216
Total	30	121	565

$\bar{x} = \frac{1}{N} \Sigma fx = \frac{121}{30} = 4.03$

$\sigma^2 x = \frac{\Sigma f x^2}{N} - \bar{x}^2 = 2.57$

$S.D. = \sqrt{\sigma^2 x} = \sqrt{2.57} = 1.60$

Mean $= 4.03$ S.D. $= 1.60$

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(4) Find the mean and the standard deviation of the first n natural numbers. Hence find the mean and the standard deviation of the first 200 natural numbers.

Solution : First n natural no.

$1, 2, 3, 4 - - - - n$

$1+2+3+ - - - +n = \sum_{r=1} r = \frac{n(n+1)}{2}$

$\therefore \bar{x} = \frac{1}{n} \times \frac{n(n+1)}{2} = \frac{n+1}{2}$

$\sum x^2 = 1+2^2+3^2+ \cdots n^2$

$\sum_{r=1}^{n} r^n = \frac{n(n+1)(2n+1)}{6}$

$\sigma^2 x = \frac{1}{n} \sum x^2 - (\bar{x})^2$

$\sigma^2 x = \frac{1}{n} \times \frac{n(n+1)(2n+1)}{6} - \frac{(n+1)^2}{2}$

$\sigma^2 x = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$

$\sigma^2 x = (n+1)\left[\frac{2n+1}{6} - \frac{n+1}{4}\right]$

$\sigma^2 x = (n+1)\left[\frac{4n+2-3n-3}{12}\right]$

$\sigma^2 = \frac{(n+1)(n-1)}{12}$

$S.D. = \sqrt{\sigma^2} = \sqrt{\frac{(n+1)(n-1)}{12}}$

Mean $\bar{x} = \frac{n+1}{2}$

Put $n = 200$

$\therefore$ Mean $\bar{x} = \frac{201}{2} = 100.5$

$S.D. = \sqrt{\frac{(n+1)(n-1)}{12}}$

Put $n = 200$

$S.D = \sqrt{\frac{201 \times 199}{12}}$

$S.D. = \sqrt{\frac{39999}{12}}$

$S.D. = \sqrt{3333.25}$

$S.D. = 57.7$

(5) The following table shows weight of students of two classes. Calculate the coefficient of variation of the two distributions. Also find the mean and the standard deviation for both the classes.

Weight in Kg.	Class A	Class B
30-40	8	9
40-50	16	12
50-60	12	18

Solution : .....

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For class A Let $u = \frac{x-45}{10}$

Wt in kg	mid value x_i	f_i	u_i	f_iu_i	f_iu_i
30-40	35	8	-1	-8	0
40-50	45	16	0	0	0
50-60	55	12	1	12	12
Total		36	0	4	20

$\Sigma f_iu_i = 4$, $\Sigma f_iu_i^2 = 20$, $N = \Sigma f_i = 36$

Now $\bar{u} = \frac{\Sigma f_iu_i}{N} = \frac{4}{36} = \frac{1}{9}$

$Var(u) = \sigma^2_u = \frac{\Sigma f_iu_i^2}{N} - (\bar{u})^2 = \frac{20}{36} - \frac{1}{81}$

$= \frac{180-4}{224} = \frac{176}{324} = 0.54$

$Var(u) = \sigma^2_u = 0.54$

$Var(x) = \sigma^2_x = h^2 Var(u) = 100 \times 0.54 = 54$

$S.D. = \sigma_x = \sqrt{54} = 7.34$

$mean = \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{35\times8 + 45 + 16 + 55 + 12}{36}$

$mean = 46.11$

$C.V. = \frac{S.D}{mean} \times 100 = \frac{7.34}{46.11} \times 100 = 5.91$

For class B Let $u = \frac{x-45}{10}$

u_i	mid value	f_i	u_i	f_iu_i	f_iu_i
30-40	35	9	-1	-9	9
40-50	45	12	0	0	0
50-60	55	18	1	18	18
Total		39	0	9	27

$\Sigma f_i = N = 39$, $\Sigma f_iu_i = 9$, $\Sigma f_iu_i^2 = 27$

Now $\bar{u} = \frac{\Sigma f_iu_i}{N} = \frac{9}{39} = 0.23$

$Var(u) = \sigma^2_u = \frac{\Sigma f_iu_i^2}{N} - \bar{u}^2 = \frac{27}{39} - (0.23)^2$

$= 0.93 - 0.05 = 0.85$

$\sigma^2_u = 0.88$

$\sigma^2_x = h^2 \times Var(x) = 100 \times 0.88 = 88$

$S.D = \sqrt{88} = 9.38$

$mean = \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{35\times5 + 45 + 12 + 55 + 18}{39}$

$mean = \frac{1845}{39} = 45.30$

$C.V. = \frac{S.D}{mean} \times 100 = \frac{9.38}{47.30} \times 100 = 19.83$

(6) The means of two samples of sizes 10 and 20 are 24 and 45 respectively and the standard deviations are 6 and 11. Obtain the standard deviation of the sample of size 30 obtained by combining the two samples.

Solution :

Given

$n_1 = 10$, $\bar{x}_1 = 24$, $\sigma_1 = 6$

$n_2 = 20$, $\bar{x}_2 = 45$, $\sigma_2 = 11$

Combined mean $= \bar{x}$

$= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$= \frac{10\times24 + 20\times45}{10+20}$

$= \frac{240 + 900}{30}$

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$$= \frac{1140}{30}$$

$$\bar{x} = 38$$

$d_1 = \bar{x}_1 - \bar{x} = 24 - 38 = -14$

$d_2 = \bar{x}_2 - \bar{x} = 45 - 38 = 7$

Combained S.D. $= \sigma = \sqrt{\frac{n_1(d_1^2 + \sigma_1^2) + n_2(d_2^2 + \sigma_2^2)}{n_1 + n_2}}$

$$= \sqrt{\frac{10(196 + 36) + 20(49 + 121)}{10 + 20}}$$

$$= \sqrt{\frac{2320 + 3400}{30}}$$

$$= \sqrt{\frac{5720}{30}}$$

$$= \sqrt{190.66}$$

$$= 13.80$$

Sign of Teacher :

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